RLMS Project

1. Team Members and Roles

- **Aouatif Guennoun:**
  - Was in charge of writing and creating the different versions of our ER designs.
  - Leader of the group and will be the one managing the GitHub repository.
  - Contributed to the realization of the final version of the PowerPoint document.
- **Valdez Kemekong:**
  - Contributed to finding solutions whenever the group struggled by giving real-world examples.
  - Helped in creating the PowerPoint slides.
- **Ranya El Bahali:**
  - Discussed with the members of the team the different possible solutions for each task.
  - Was in charge of writing the report and identifying the issues faced during this project.
  - Provided the material needed to work on the assignment.
- **Hayat Ennani:**
  - Shared with the team members different suggestions about each task and the most efficient way to solve them.
  - Participated in creating the project report and provided the last version of it.

2. Key Challenges & How They Were Addressed

- **Key and participation constraints:** Translating the sentences into symbols was an issue that the team faced during this project. This was dealt with by carefully understanding each task and the relationships between entities.
- **Expressing exceptions:** In the ER diagram, some relationships required certain categories of entities in the same entity set, which made the realization of the design difficult to express. This led us to explain it in more detail in the next section.
- **The ambiguity of the sentences:** Some tasks were difficult to understand as they mentioned multiple relationships between different entities, which made us a bit confused about whether we should choose a binary or a ternary relationship, especially with multiple constraints.
- **ISA:** In some cases, an attribute can have multiple categories, which made it difficult for us to decide whether this attribute should become an entity or remain the same. Based on these categories, if they have attributes that will inherit new ones from the superclass, then ISA was the right solution.
- **Conditions:** In some relationships, a condition should be verified for the relation to occur, which was hard to express in the ER design. Therefore, in some tasks, we just assumed that the whole entity set verified the condition in question.

3. Modeling Choices & Justifications

- **PE3:** The hypothesis in this task was made because we did not know whether a student could be enrolled in different programs or only one, so we assumed that they could attend multiple programs.
- **PE5:** As discussed in the previous section, regarding the report relationship, all employees report to only one employee except the dean, and we could not find a way to express it in the ER design. Since the relation allows an employee not to participate in a relation, the dean is free to not report to anyone
- **EQ4:** In this task, we cannot distinguish whether the relation keeps only the recent location or saves the location history.
- **EQ6:** We added a new attribute to the equipment unit, which is portability, so that we can differentiate between fixed equipment and portable ones.
- **EQ7:** In this task, the relationship only targets eligible people, so we assumed that the whole entity set was eligible.We assumed that the represented entities and relationships gives enough information

to model the condition later in a database

- **CA4:** We added new attributes to the relationship *held by*, such as record issue date, expiration date, and result, so that we can identify each person's certification.
- **CE7:** Since the condition can't be clearly represented, we assumed that the represented entities and relationships gives enough information to model it later in a database.
- **RE1:** We had a relationship between reservation and person, and between research project and equipment unit. Based on the constraint, we decided to have a ternary relationship between reservation, person, and research project, and a binary one between equipment unit and reservation. We could not, once again, target all the entity set except administrative staff.
- **RE3:** As before, in this task, we supposed that all people are designated by the dean, so we consider them as approvers.
- **MA2:** According to the text given, maintenance is considered a weak entity, as it is identified by the pair (serial number, start time).
- **MA3:** We assumed that technical staff can carry out multiple maintenance operations on one equipment unit, and we decided to have a ternary relationship as the constraint did not hinder that.
- **CA3:** We assumed that an equipment unit can have multiple calibration records, as nothing was mentioned in the text.
- **CA4:** Calibration records are identified by a pair that involves the calibration date and the ID of the equipment unit.
- **CO4:** In this task, we based our design on a binary relationship between consumable, laboratory, and suppliers, because the relation between consumable and laboratory had some attributes, which made the ternary relationship harder to work with in this case.

4. Requirements Not Expressible in the ER Diagram

As discussed in the previous sections, conditions, exceptions, and targeting a category of an entity set could not be expressed in the ER design, which is the case for PE5, PR2, PR3, CA7, RA1, and RA3. Therefore, we decided to clarify this in this section.